



Sign and sound in the study of written texts

1. The increased interest in speech brought about by the development of modern Linguistics has resulted in a corresponding diminution of regard for written texts as a manifestation of language. In fact a majority of linguists, at least in North America, appear to regard only speech as "real language," and writing as an incompetent attempt to record it, giving "an imperfect and incomplete picture of actual speech."¹ The purpose of this paper is to show that this attitude reflects a distorted view of the relation of speech and writing, and has had a pernicious effect on the grammatical study of written texts.

2. American linguists since Bloomfield appear to have defined language as spoken.² The admissibility of such a definition, (philologically impeccable, if somewhat narrow in contrast to current usage) cannot be denied. Its use, however, requires some term other than "language" for the system of items and relations represented by the sounds of speech. The term **grammar** is used here for this purpose. It comprehends morphemes, their arrangements (morphology and syntax) and their meanings (semantics).³ "Speech" here refers to a spoken utterance, "writing" to a written text, each, in its way, representative of the **grammar**. To state

¹H. A. Gleason, *An Introduction to Descriptive Linguistics* (New York, 1965) 1.12 (p. 10).

²So L. Bloomfield, *Language* (London, 1935), p. 21. It is interesting that Sapir, *Language* (New York, 1921), p. 7, does not include 'vocal' in his definition of language. Many European linguists would not do so either. Firth appears to reach the other extreme in defining 'speech' as 'oral or written', in "The Semantics of Linguistic Science," *Papers in Linguistics* (London, 1957), p. 144.

³From the bareness of this statement it can be seen that no particular theory of language is presupposed here. However, the distinction between grammatical and lexical form, or between a form and its meaning, and the classing of the units so distinguished under separate terms, seems an unnecessary complication, at least for the practical study of language.

suffixes are not a characteristic feature of all OE adjectives, just as a positive suffix is not a characteristic feature of all OE adverbs. A second reason is that inflectional suffixes—such as those of the OE adjective strong/weak paradigms—usually have a connection with sentence concord, but the suffixes of comparison in OE adjective and adverb sets do not. Hence, for these two basic reasons the suffixes of comparison are labeled herein as postbases rather than as inflectional suffixes.

On the basis of the evidence presented, it is apparent that the OE morphological set herein called adverb consists of subsets with positive, comparative, and superlative constructions marked by characteristic postbases, and that the set as a whole is an uninflected set. In contrast, the OE morphological set herein called adjective constitutes an inflected morphological set of subsets with positive, comparative, and superlative constructions, the last two of which are marked by characteristic postbases, and added to all three in sequential order are inflectional suffixes of the strong/weak adjective paradigms. This information may be summarized as follows:

OE adverb = base + PoB (1, 2, 3)

OE adjective = base + PoB (2, 3) + strong/weak suffix

Thus, numerous OE constructions, as for example OE *fela* 'much', *gēara* 'formerly,' and other traditionally labeled OE adverbs on the basis of syntactic function, would not qualify as morphological adverbs on the basis of the description given herein.

Although the matter of syntactic classes in OE is properly outside the subject of this paper, nevertheless, so-called OE adverbs that do not qualify as morphological adverbs should be relabeled as adverbials, thus descriptively positing the necessary distinction between morphological adverbs and syntactic adverbials.

the case in this way is no doubt indicative of the author's bias (just as the definition of language as spoken is indicative of bias), but this statement should be justified in what follows.

3.1. The most obvious question raised by the above statement concerns the relationship of speech to the **grammar**. Many linguists state that "morphemes are composed of phonemes." Their own works often show this position to be untenable, suggesting that it is a badge of faith in speech as "real language" rather than a carefully thought out proposition.⁴ If it is true, how are the different allomorphs ("composed" of different sets of phonemes) to be related to the morpheme? A change to "morphemes are stateable in terms of phonemes"⁵ amounts to agreement with the position taken here. Morphemes are not composed of phonemes, but are represented by them. Spoken sounds are not part of the **grammar** of a language, but a means of representing it.

3.2. It is necessary, as often in linguistic discussion, to remember what a phoneme is. It is not a sound, a "fact of language," it is a classification or description of sounds. There will be various possible ways of making such a classification for any spoken language, all equally valid. The particular classification used, and indeed the definition of "phoneme" envisaged, will depend on the interests of the scholar, and the purpose for which the phonemic analysis is being made.⁶ A morpheme, then, cannot possibly be "composed of" phonemes. The range of sounds used in any spoken language is characteristic of that language, as is the way in which these sounds are used to represent morphemes. However, the sounds used in a language may vary widely in different localities, as is the case in English, but such variations in the sound system are accompanied only by minor, and not necessarily correlative, changes in the **grammar**.

⁴E.g., Harris, in *Methods in Structural Linguistics* (Chicago, 1951), speaks of morphemic segments as "sequences of phonemes," and phonemes "constituting" a morpheme (pp. 195-6 passim), yet on p. 184 he discusses order as a morphemic element. Such an element can only be "represented by" phonemes, it cannot "consist of" them. (See also note 5.)

⁵H. A. Gleason, Introduction, p. 75, following "Morphemes are generally short sequences of phonemes" and similar statements (pp. 51-3).

⁶E.g., the material on the basis of which a phoneme is determined may range from a single idiolect, as in D. Jones, *The Phoneme* (Cambridge, 1962), Chapter 2, (note especially #28, p. 9) to a wide group of dialects, as in Trager and Smith, *An Outline of English Structure* (Studies in Linguistics, Occasional Papers 3, Washington, 1957), Section 1.

There is obviously, then, no indissoluble bond between phonemes and morphemes. A morpheme does not consist of phonemes, it is represented by them, or, to be more accurate, by sounds which, for purposes of linguistic description, are classified as allophones of phonemes.

4. If morphemes are only represented by, not composed of, phonemes, the question of whether the sounds of speech always indicate all the morphs of the grammatical structure they represent may be raised. The interest of the Firthian school in the "context of situation" is an indication that they do not. The existence of "zero morphs" and "homophones" strongly suggests that they do not. Features coming under Chomsky's heading of "deeper structure" can also be interpreted in this way. In any case coming under these headings, the normal speaker of the language will "understand" some grammatical elements in addition to those actually represented in sound. It is presumably possible to explain most such cases of "understanding" by reference to complete patterns of the **grammar**: e.g., paradigmatic patterns in the case of "zero morphs" or collocations in the case of homophones. The fact remains, however, that these features are not represented in sound in a particular utterance, their existence can only be deduced from comparison of a number of utterances. Speech does not, then, always indicate all the morphs of the structure it represents.

5. The conclusions about the relation of speech to the **grammar** can, then, be stated as follows: The sound system of a spoken language is characteristic of that language. The elements of the **grammar** are not, however, formed from sounds, but represented by them. The representation may be complete or incomplete according to the situation. The concept of "phoneme" is a tool designed to permit the simple and clear description of the way in which this representation is achieved.

6.1. Speech, then, is the representation of the **grammar** of a language in sound. Writing, according to many linguists, is an imperfect representation of speech. The "imperfection" which they see in traditional writing systems consists in their falling short of the "ideal" of phonemic transcription.⁷ Granted their

⁷Phonemic transcription also, of course, gives an "imperfect and incomplete picture of actual speech" (see note 1), a fact not noticed by those who condemn traditional writing systems on this ground. The linguist will claim that all "significant" features of speech are recorded in phonemic transcription—but

point of view, they are right. No traditional writing system appears ever to have approached adequacy as a phonemic transcription. Our own system, when it originated in the second millennium B.C., did not even represent vowel sounds.⁸ During its history it has been modified to suit the purposes of various users, but never so far as to deserve the term "phonemic." There are, of course, exceptions to this statement in some writing systems devised by individuals,⁹ but none of these ever came into general use. Such individual systems, however, show that the concept of phonemes, although not labelled, was available to the ancients, and that they had the technical ability to produce a phonemic transcription.¹⁰

6.2. If a phonemic transcription were the ideal writing system, and could have been produced, it is rather surprising that it never was brought into use. In fact, of course, a phonemic transcription is a specialized tool made for a specialized purpose. It is quite unsuited to the requirements of a general writing system, being far more complex than is necessary or desirable.¹¹ Its use as a general writing system would also be severely restricted in space and time.¹²

significant to whom and for what? (see note 6). Other features of speech, such as the "tones of voice" expressing anger, admiration, boredom, etc., are certainly significant to the hearer, although generally classed as "paralinguistic," i.e., beyond the linguist's concern.

⁸This is true whether one holds the traditional view that it was genuinely alphabetic, or Gelb's view in *A Study of Writing* (London, 1952), p. 147f., that it was originally a syllabic system.

⁹E.g., Steele and other English scholars discussed by D. Abercrombie in *Studies in Phonetics and Linguistics* (London, 1965), Chapters 5 and 6.

¹⁰Note also, besides the Indian tradition represented by Pāṇini, the description of the vowels of Hebrew according to their position of articulation by Saadya Ga'on (d. 942) given in S. L. Skoss, *Saadia Gaon, the Earliest Hebrew Grammarian*, (Philadelphia, 1955) p. 27f., and the description of the consonants of Arabic on the same basis by Ibn Sīna (Avicenna, d. 1037), which is preceded by a detailed description of the physical production of speech sounds, given in K. I. Semaan, *Arabic Phonetics* (Lahore, 1963).

¹¹Even a child knows the intonation contours typical of his language before he begins reading. It is therefore unnecessary to mark them. It is interesting, in this connection, to compare Pitman's "Initial Teaching Alphabet" (a semi-phonetic spelling designed for the teaching of beginning readers, and used with success in the U.K. and the U.S.A.) with a phonemic transcription, from which it differs in a variety of ways.

¹²Contemporary spoken English differs dialectically in the number of phonemes and their use, as is shown for example, in the section on phonology in Trager and Smith's *Outline of English Structure*, especially Sect. 1.33, p. 22f. Sound changes will induce similar differences in a given locality over a period of time. If phonemic transcription were used as a general writing system, several systems would be needed, no matter how broad a definition of

7.1. If writing is not an attempt at phonemic transcription, what, then, is it? R. H. Robins says "It is broadly true that one prime purpose of any writing system is to represent in permanent form the distinctive sound units or phonemes of a language," and takes some pains to support this claim.¹³ It is not, however, really true to say that "In spellings like *rough* and *cough*, *ou* and *gh* indicate /ʌ/ and /ɔ/, and /f/." Because of the diversity of speech patterns within a language, a given spelling will "indicate" different sounds to different readers. The reason for this is that the sounds "indicated" are not attainable directly from the written form, but only by way of the morph. *Rough* "indicates" /r ʌ f/ to Robins only because he recognizes the morph represented by the letters, and knows the conventional representation of that morph in speech. The letters themselves do not preclude other pronunciations.¹⁴

7.2. What writing does, then, is represent the morphemes of a language sufficiently well to permit their recognition by the reader trained to them.¹⁵ This is obvious for "ideographic" writing systems, where each sign represents a morph or group of morphs. Such a system has its advantages, among them complete independence of dialect, or even language, boundaries.¹⁶ Their obvious disadvantage is the number of signs required.¹⁷ The

"phoneme" was envisaged, even in North America, (cf. Gleason on the transcription used in the "English-for-Foreigners" programme of the American Council of Learned Societies [in Introduction, 19.17-18, p. 324 f.]), let alone the rest of the English speaking world. Furthermore, these systems would need periodic revision.

¹³In CJL 9, (1964), p. 108 f.

¹⁴Of course broad general rules for the determination of spoken forms from written forms can be given, and these often enable readers to recognize a morph, when they are familiar with its spoken, but not its written representation. The traditional "spelling out" method of teaching reading is based on this. These rules, however, cannot be systematized. See note 15.

¹⁵The reader will, of course, normally state his recognition in terms of the spoken representative of the morph. However, because the connection between written and spoken form is only indirect, it is possible to "mis-recognize" a written form, even when the spoken counterpart is known, and even to the extent of including a new morph in the idiolect. E.g., *mised* is not uncommonly ascribed to a verb *mizle* or *mizzle* rather than *mislead*.

¹⁶Cf. the history of the Chinese writing system. The "Arabic" numerals are also a case in point.

¹⁷Thus the teaching of beginning readers by the "word recognition" method is unsatisfactory not because the theory on which it is based is wrong, but because there is no way in which the student can proceed on his own after acquiring basic principles.

alphabetic system avoids this problem by using only a small number of signs. Each morph, however, is (most commonly) represented by a group of signs, instead of only one. This is the same type of method as that used in speech for the representation of morphs by phonemes. An alphabet, then, is an imitation of the phoneme system of a language, although, as we have seen, it is not a representation of it.¹⁸ Even in an alphabetic system, however, the "ideographic" aspect of writing remains,¹⁹ and is maintained by the common use of symbols simpler than the traditional spelling to represent common morphs or groups of morphs.²⁰

8. The necessary conclusion, then, would seem to be that the morpheme system of a language, like the syntactic and semantic systems, does not exist in physical form. Either phonemes or graphemes may be used to represent it, but morphemes are not composed either of phonemes or of graphemes. The grapheme system is an imitation of the phoneme system, not an inefficient attempt to represent it. Graphemes and phonemes thus form independent systems of representing the **grammar**, used on the same level.²¹ Thus this paper is in accord with Bloomfield's dictum "Writing is not language, but a means of recording language by visible marks,"²² if it be conceded that he means by "language" what is here meant by **grammar**. We can then add as corollary, "Speech is not **grammar**, but a means of representing **grammar** by audible sounds."

9.1. Those who think only of speech as "real language" use various arguments to justify their exclusion of written texts. Some, such as the common statement that speech is both relatively and absolutely a precursor of writing, are quite irrelevant. Others, such as the charge that writing uses superfluous

¹⁸This can be easily seen in the use of "punctuation marks." E.g., the question mark can be said to be used in imitation of the rising inflection of speech. Its use would coincide with rising inflection in speech in a considerable proportion of cases, but by no means in all.

¹⁹Noted first, perhaps, by de Saussure, *Cours de linguistique générale* (Paris, 1922), p. 57.

²⁰E.g., the many "abbreviations" of mediaeval MSS such as *Dno* for *Domino*, and writings such as *U.S. \$10.50* today. The convention that "letters represent sounds" is strong enough to produce spoken counterparts for a great many (e.g., *N.A.T.O./neytow/*), even where the combination of letters is impossible in English (e.g., *N.F.C.U.S./nifkəs/*).

²¹This is not a new suggestion, see, for example, McIntosh in "The Analysis of Written Middle English," *TPS* 1956, p. 40.

²²Language, p. 21.

symbols, apply equally well to speech.²³ The claim that "writing is a secondary visual representation of speech" is not true in the sense intended. Writing does not represent the morpheme system only by way of the sound system. As we have seen it would be much more true to reverse this statement. Consequently, any argument that writing fails to represent many features of speech is true but irrelevant. It is also true that writing sometimes fails to represent features of the **grammar** that are represented in speech.²⁴ However, the reverse is also true.²⁵ Writing cannot, therefore, be rejected as an object of linguistic study on the grounds that it gives an inadequate representation of spoken language.

9.2. Some maintain that written language is distorted, governed by rules which are valueless because not valid for the "real language" of the spoken form, and pernicious because there is a tendency to hold them up as a standard for it. "Nobody ever made a grammatical error in a non-literate society"²⁶ is a statement typical of this viewpoint. This is, of course, simply a fragment of linguistic mythology. Children make innumerable mistakes in learning their mother tongue. Frequently these are caused by unconscious recognition of linguistic patterns: the more common form of a morpheme is used where an uncommon allomorph is customary. Sometimes such "mistakes" are not dropped from childhood speech, but maintained, eventually causing the replacement of the rarer by the more common allomorph throughout the language. Such analogic replacement is a common cause of linguistic change.

9.3. Whether or no during the course of such a change either form would be considered an error depends on whether the speakers recognize an ideal standard for their language, and how firmly they maintain it. This is a matter of social custom, not of linguistic medium. Such a standard may be based on a written

²³Symbols such as *c*, *q*, and the capital letters have their own place in the conventional written representation of English (for capitals, see note 25). They cannot be considered superfluous any more than allophones can be considered superfluous sounds.

²⁴E.g., *export* can represent a noun /'ekspɔnt/ or a verb /ek'spɔnt/.

²⁵E.g., in such pairs as *tail*: *tale*, *Liberal*: *liberal*. Cf. also #4.

²⁶H. M. MacLuhan, *The Gutenberg Galaxy* (Toronto, 1962), p. 238, coupled with his even more surprising assertion that "the printing press made bad grammar possible" (p. 231); cf. note 27.

form of the language,²⁷ but this is not necessarily the case. In England the speech of the "upper classes" used to form a standard.²⁸ This could not possibly have been derived from written English. Among the Arabs, the speech of the Bedouin was looked on as ideal, so much so, apparently, that the language of the Qur'ān, which later became the standard, was remodelled to conform to it.²⁹ Even where some specific standard of this sort is not maintained, it seems to be typical of human behaviour for the members of a community to feel that their speech forms are ideal, and that any differences shown by their neighbours are errors. This was true in ancient Greece³⁰ as it is in modern America. Such community speech forms are rarely based on a written ideal.³¹ It may be difficult to find an example of an ideal standard of this sort from a genuinely non-literate society, but it would be surprising if it were impossible.

9.4. It is perfectly true that the written form of a language may use structures foreign to its spoken counterpart. This is not an indication that writing is bound by an archaic complex of grotesque and useless conventions, but merely a fact of social usage. The history of the written form—that it changes more slowly than the spoken, or that it has been moulded by theoretical rules of "correctness"—can be of no significance to a synchronic

²⁷The language of the Bible has long formed the standard for Hebrew. The earliest record comes from Saadya Ga'on (d. 942): "I likewise saw a number of Israelites who do not observe the simplest rules of our language, not to mention the more difficult ones, and they mispronounce it in their speech. . . ." (quoted in Skoss, Saadya Ga'on, p. 58).

²⁸This standard is now less admired than formerly. It is well illustrated by the writings of G. B. Shaw. See a collection, including excerpts from "Pygmalion," the most obvious example, in A. Tauber, ed., G. B. Shaw on Language, (London, 1965).

²⁹See P. E. Kahle, *The Cairo Geniza* (Oxford, 1959), p. 141 f., and references there.

³⁰See, for example, Aristophanes' fragment 198, on teenage slang, and 685, distinguishing city speech (effeminate) and country speech (vulgar) from "average Attic," in J. M. Edmonds, *The Fragments of Attic Comedy*, vol. 1 (Leiden, 1957).

³¹Examples can be found in T. Pyles, *Words and Ways of American English* (London, 1954), Chapter 11. His remarks show that in many (most?) cases the condemnation of a particular structure stems from a theoretic ideal, not actually adhered to in the written language. An example of a community speech form contravening the written ideal occurs in the story quoted from R. A. Hall (p. 204), showing a case where a man using "upper class" speech forms was not fully admitted to a community until he adopted its "lower class" forms. This sort of thing is, of course, paralleled by the necessity of using the "correct" slang in various groups, of which teenagers are a standard example.

study, and can only add to the interest of a historical one. To exclude writing from consideration as a manifestation of language on the basis of these and similar factors is pure prejudice. It is well known that, in many societies, including our own, different forms of spoken language are required in different social situations. Why should the written form be treated as though it were on a different level?³² The spheres of use of written and spoken language are largely separate. It is to be expected that different forms of language should be used. There is no reason whatever to regard only one of these forms as "real language."

10.1. There are no grounds, then, for rejecting writing as a manifestation of language. Nevertheless, there is a deplorable tendency on the part of linguists, when studying written texts, to convert them from their original writing system into a so-called phonemic transcription, and so to pretend to be studying records of speech.³³ It is difficult to see how this practice could be justified, even where the scholar is familiar with the phonemic system used, for, as we have seen, writing and speech frequently follow different conventions. Where the phoneme system is not well known, as will normally be the case, the method is indefensible. An interpretation in terms of sound of a set of signs which are themselves at best inexactly and probably inconsistently imitating a set of sounds must be largely subjective. Various quite contradictory schemes can be drawn up, each of which satisfies the available facts equally well,³⁴ and choice among

³²Written language, of course, is also used in different forms in different situations: e.g., legal documents, newspaper reports, intimate letters.

³³An extreme example is G. M. Schramm, *The Graphemes of Tiberian Hebrew* (Berkeley, 1964), in which the graphemes, the supposed subject of the study, are transcribed (not transliterated, p. 32, note 1) into sound symbols before their distribution is analysed. Important features of the writing system, such as the "shewa sign" and word final "alef" and "he" are not marked in the transcription, and their distribution therefore not analysed, so that Schramm cannot be said to be analysing the writing system. The sound symbols used are not intended to represent phonemes (p. 32), so it would be difficult to say just what he is analysing, and it is difficult to see how anything but gain could have resulted from analysing the distribution of the graphemes themselves, and basing the phonological transcription on this information. There are, however, some notable exceptions to this tendency to treat writing as an intended record of speech, e.g., M. A. K. Halliday, "Secret History of the Mongols," *The Language of the Chinese* (Oxford, 1959).

³⁴Opinions on the significance of the standard signs for the vowels of Biblical Hebrew ranges from "basically phonemic" (so S. Morag, *The Vocalization Systems of Arabic, Hebrew, and Aramaic* [The Hague, 1962], p. 29) to

them can only be arbitrary. Analysis based on such criteria would describe a language which cannot be shown to have existed. The writing system, on the other hand, is designed to indicate the morphemes of the language to the reader. Subjective decisions may be needed here too, but the results of error are hardly likely to pervade the whole analysis, as would be the case with error in phonemic interpretation. Obviously in such cases the writing system provides the firmer basis for grammatical study.

10.2. It is, of course, necessary to interpret writing in terms of sound for purposes of historical study of the sound system of a language. Even here, however, the basis for analysis must be a thorough study of the writing system. If the **grammar** is the subject of study, the analysis must be carried out in terms of written forms. Only in this way can the maximum of unquestionable information about the **grammar** be obtained. It is quite as unsuitable for a written text to be analyzed in terms of sound as for a spoken text to be analyzed in terms of writing.

11. The conclusion of this paper is, then, that there is no ground for the rejection of writing as a manifestation of language, and further, that techniques developed for the study of speech must not be used unchanged in the study of written texts. It is not intended at all to disparage these techniques when used in their proper sphere. The intention is to justify the study of writing as a representative of the **grammar** of a language parallel to, not dependent on, the spoken form of that language, and to encourage the development of techniques for the study of writing parallel to those developed for the study of speech.

"basically phonetic" (so C. Rabin, "The 'Small Vowels' in Tiberian Hebrew," in *Studies in Honour of N. H. Tur-Sinai* [Publications of the Israel Society for Biblical Research 8, Jerusalem 1960, p. 206]. There is a correspondingly wide variety of phonemic interpretations of the system.